BMHS MA0 Tournament - 2018 (compliments of mr. vu)

Calculus Individual Test - BC

1) Find the y-value of the absolute maximum for the function $y = -(x+6)^{\frac{2}{3}}$ on the interval $[-4, 1]$

2) Find the equation of the line tangent to the graph of $y = \dfrac{1}{5x^4 + 5}$ at the point $\left(1, \dfrac{1}{10}\right)$.

3) Use implicit differentiation to find $\dfrac{dy}{dx}$ at $(2, -1)$.

$5 = 5x^2 + 3x^2y + 3y$

4) Find $\displaystyle\int_0^2 \frac{24x}{(4x^2+4)^2}\,dx$

5) Find the area of the region enclosed by the curves.

$y = 3\sqrt{x},\ \ y = -3\sqrt{x},$
$x = 0,\ \ x = 4$

6) A spherical snowball melts at a rate of $\dfrac{32\pi}{3}$ in³/sec. At what rate is the radius of the snowball changing when the radius is 4 in? Include units.

7) Approximate the area under the curve over the interval $[-2, 2]$ using 2 midpoint rectangles.

$y = x^2 + x + 1$

8) Find $\displaystyle\lim_{x\to 0} \frac{2x}{\ln(x+1)}$

9) If $F(x) = \int_{x}^{2x} (t^3 - 2t^2 + 1)\, dt$ then find and simplify $F'(x)$

0) For the function $y = 2x^2 - 12x + 17$, where does the instantaneous rate of change equal the average rate of change on the interval [2, 5]

11) Find the volume of the solid that results when the region enclosed by the curves is revolved about the x-axis.

$y = \sqrt[3]{x},\ y = 0,\ x = 1$

12) Find $\int \frac{\ln x}{\sqrt{x}}\, dx$

13) Consider the curve defined on the interval $[\,0, 2\pi\,)$ by the parametric equations

$x(t) = \cos 3t$
$y(t) = 3\sin 2t$

Find the slope of the line tangent to the curve when $t = \frac{\pi}{4}$

14) Find the Average Value of the function over the interval [2 , 5]

15) If a particle's position is defined by the vector valued function $< \cos 3t\ ,\ 6t^4 - e^{2t} >$ then find the particle's acceleration vector when $t = 0$

Brother Martin Solutions

1. $-\sqrt[3]{4}$
2. $y = -\frac{1}{5}x + \frac{3}{10}$
3. $-\frac{8}{15}$
4. $\frac{3}{5}$
5. 32
6. $-\frac{1}{6}$ inches per second
7. 8
8. 2
9. $15x^3 - 14x^2 + 1$
10. $\frac{7}{2}$
11. $\frac{3\pi}{5}$
12. $2\ln x \cdot \sqrt{x} - 4\sqrt{x} + C$
13. 0
14. $\frac{4}{3}\ln(\frac{5}{2})$
15. $< -9. - 4 >$